

Data Dictionary

Customer Orders Data Model

Source File: customer_orders.csv

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Schema Type: Star Schema

Tables: 3 (1 Fact Table | 2 Dimension Tables)

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1. Overview

This data dictionary documents the **Customer Orders** SQL database, which is structured as a star schema comprising one fact table (fct_orders) and two dimension tables (dim_customers and dim_products). The model was derived from

raw order transaction data and normalized to remove redundancy, enforce data integrity, and support analytical queries.

The tables below summarize the tables included in this data model, their types, approximate row counts, and a brief description of each table's purpose.

Table Name	Table Type	Row Count	Description
dim_customers	Dimension	6	Deduplicated customer reference data
dim_products	Dimension	8	Deduplicated product catalog with category
fct_orders	Fact	10	One row per order transaction

2. Entity Relationship Summary

The Customer Orders data model follows a classic **star schema** pattern. The central fact table, `fct_orders`, stores one row per order transaction and carries two foreign key references that link it to its surrounding dimension tables:

- `fct_orders` links to `dim_customers` via **customer_id** (many-to-one relationship)
- `fct_orders` links to `dim_products` via **product_id** (many-to-one relationship)
- A single customer can have **many orders**; a single product can appear in **many orders**

Note: All foreign key constraints are enforced at the database level via `CONSTRAINT` declarations in the `fct_orders` DDL. Referential integrity is guaranteed — no orphaned order records can exist.

From Table	From Column	Relationship	To Table	To Column
fct_orders	customer_id	Many-to-One	dim_customers	customer_id
fct_orders	product_id	Many-to-One	dim_products	product_id

3. Table — dim_customers

Type: **Dimension** | Row Count: 6

Description

Stores unique customer reference data. Each row represents one distinct customer, deduplicated from the raw order data. This table serves as the authoritative source for customer identity within the model.

DDL

```
CREATE TABLE dim_customers (
  customer_id INT PRIMARY KEY,
  customer_name VARCHAR(100) NOT NULL,
  customer_email VARCHAR(150) NOT NULL UNIQUE );
```

Column Definitions

Column Name	Data Type	Constraints	Nullable	Description	Example Value
customer_id	INT	PRIMARY KEY	No	Unique numeric identifier for each customer	1, 2, 3
customer_name	VARCHAR(100)	NOT NULL	No	Full name of the customer	Alice Johnson
customer_email	VARCHAR(150)	NOT NULL, UNIQUE	No	Unique email address used to identify customers across orders	alice.j@email.com

Sample Data

customer_id	customer_name	customer_email
1	Alice Johnson	alice.j@email.com
2	Bob Smith	bob.s@email.com
3	Charlie Brown	charlie.b@email.com
4	Diana Prince	diana.p@email.com
5	Eve Cooper	eve.c@email.com
6	Frank Castle	frank.c@email.com

4. Table — dim_products

Type: **Dimension** | Row Count: 8

Description

Stores unique product catalog data. Each row represents one distinct product, deduplicated from raw order data and categorized by product type. This table provides descriptive attributes used to filter and group orders in analytical queries.

DDL

```
CREATE TABLE dim_products (
  product_id INT PRIMARY KEY,
  product_name VARCHAR(100) NOT NULL,
  product_category VARCHAR(50) NOT NULL );
```

Column Definitions

Column Name	Data Type	Constraint s	Nullable	Descriptio n	Example Value
product_id	INT	PRIMARY KEY	No	Unique numeric identifier for each product	101, 102, 103
product_name	VARCHAR(100)	NOT NULL	No	Name of the product as	Laptop, Mouse

Column Name	Data Type	Constraints	Nullable	Description	Example Value
				listed in orders	
product_category	VARCHAR(50)	NOT NULL	No	High-level grouping of the product. Accepted values: <i>Electronics</i> , <i>Furniture</i> , <i>Office Supplies</i>	Electronics

Accepted Values — product_category

- **Electronics**
- **Furniture**
- **Office Supplies**

Sample Data

product_id	product_name	product_category
101	Laptop	Electronics
102	Mouse	Electronics
103	Desk Chair	Furniture
104	Monitor	Electronics
105	Notebook	Office Supplies
106	Headphones	Electronics
107	Paper	Office Supplies
108	Desk Lamp	Furniture

5. Table — fct_orders

Type: **Fact** | Row Count: 10 | Grain: One row per order transaction

Description

Central fact table of the star schema. Each row represents one unique order transaction. This table holds the measurable business event (order_amount) and references customer and product dimension tables via foreign keys. **Grain:** one row per order.

DDL

```
CREATE TABLE fct_orders (
    order_id INT PRIMARY KEY,
    customer_id INT NOT NULL,
    product_id INT NOT NULL,
    order_amount DECIMAL(10, 2) NOT NULL,
    order_currency CHAR(3) NOT NULL DEFAULT 'USD',
    order_date DATE NOT NULL,
    order_status VARCHAR(20) NOT NULL CHECK
    (order_status IN ('delivered', 'shipped', 'processing', 'cancelled')),
    CONSTRAINT fk_orders_customer FOREIGN KEY (customer_id) REFERENCES
    dim_customers(customer_id),
    CONSTRAINT fk_orders_product FOREIGN KEY
    (product_id) REFERENCES dim_products(product_id) );
```

Column Definitions

Column Name	Data Type	Constraints	Nullable	Description	Example Value
order_id	INT	PRIMARY KEY	No	Unique numeric identifier for each order transaction	1001, 1002
customer_id	INT	NOT NULL, FK → dim_customers	No	References the customer who placed the order	1, 4
product_id	INT	NOT NULL, FK → dim_products	No	References the product included in the order	101, 103
order_amount	DECIMAL(10, 2)	NOT NULL	No	Monetary value of the order in the stated currency	1200.00, 25.50
order_currency	CHAR(3)	NOT NULL, DEFAULT 'USD'	No	ISO 4217 currency code. Defaults to USD if not provided	USD

Column Name	Data Type	Constraint s	Nullable	Description	Example Value
order_date	DATE	NOT NULL	No	Calendar date on which the order was placed (YYYY-MM-DD format)	2024-10-26
order_status	VARCHAR(20)	NOT NULL, CHECK constraint	No	Current fulfillment status of the order. See accepted values below	delivered

Accepted Values — order_status

Value	Description
delivered	Order has been successfully delivered to the customer
shipped	Order has been dispatched but not yet delivered
processing	Order is being prepared and has not yet shipped
cancelled	Order was cancelled and will not be fulfilled

Constraints Summary

Constraint Name	Type	Column(s)	References / Notes
PRIMARY KEY	PK	order_id	—
fk_orders_customer	Foreign Key	customer_id	dim_customers(customer_id)
fk_orders_product	Foreign Key	product_id	dim_products(product_id)
CHECK order_status	Check	order_status	'delivered', 'shipped', 'processing', 'cancelled'

Constraint Name	Type	Column(s)	References / Notes
DEFAULT order_currency	Default	order_currency	'USD'

Sample Data (First 5 Rows)

order_id	customer_id	product_id	order_amount	order_currency	order_date	order_status
1001	1	101	1200.00	USD	2024-10-26	delivered
1002	2	102	25.50	USD	2024-10-27	shipped
1003	3	103	150.75	USD	2024-10-28	processing
1004	1	104	300.00	USD	2024-10-28	delivered
1005	4	105	5.99	USD	2024-10-29	delivered

6. Glossary

The following table defines key technical terms used throughout this data dictionary.

Term	Definition
Primary Key	A column or set of columns that uniquely identifies each row in a table
Foreign Key	A column that references the primary key of another table, enforcing referential integrity
NOT NULL	A constraint that prevents null/empty values from being stored in a column
UNIQUE	A constraint ensuring all values in a column are distinct across all rows

Term	Definition
DEFAULT	A value automatically assigned to a column if no value is explicitly provided at insert time
CHECK	A constraint that limits which values are accepted in a column to a defined set or range
Fact Table	A table that stores measurable, quantitative business events (e.g., orders, transactions)
Dimension Table	A table that stores descriptive attributes used to filter, group, or label facts in analysis
Star Schema	A database design pattern with a central fact table surrounded by related dimension tables
Grain	The level of detail each row in a fact table represents (e.g., one row per order)
Nullable	Indicates whether a column can contain a NULL (empty/missing) value
VARCHAR	Variable-length character string data type; stores up to a defined maximum number of characters
DECIMAL	Numeric data type with fixed precision and scale; commonly used for monetary values
CHAR	Fixed-length character string data type; pads shorter values with spaces to reach the defined length
DATE	Data type representing a calendar date in YYYY-MM-DD format